Capstone Project Concept Note and Implementation Plan

Project Title: Problematic Gaming Early Warning: Does Early Gameplay Telemetry Add Predictive Value Beyond the Baseline Gaming Disorder Test Score?

Team Members

Yunus Berkay İnci

Deniz Caman

Fatih Yılmaz

İsa Adahan Ünal

Alperen Acar

Group 7 · SIC Capstone

Concept Note

1. Project Overview

Problematic gaming, characterized by impaired control, priority given to gaming over other activities, and continuation despite negative consequences, is recognized by the World Health Organization (2020) as a condition that can seriously disrupt work, school, sleep, and social life. This capstone supports Sustainable Development Goal 3 (Good Health and Well-being), in particular target 3.4 on promoting mental health and well-being, and indirectly SDG 4 (Quality Education), because disrupted routines and school or work functioning are among the first areas affected.

The problem we address is that current tools rely on self-reported or total playtime, which is a weak and noisy indicator: self-reports and logged use agree only moderately (Parry et al., 2021), and time alone says little about harm (Vuorre et al., 2022). The project asks whether gaming behavior observed in the first 14 days (timing, irregularity, trend, session structure) adds measurable predictive value for later Gaming Disorder Test (GDT) scores beyond the baseline GDT and total playtime. The potential impact is twofold: if a contribution is found, an explainable early-warning signal can be built for research and future prevention work; if not, the result shows that a short scale is sufficient and invasive behavioral monitoring is unnecessary. Both outcomes are informative. The project is a prediction study on symptom scales; it makes no clinical diagnosis claims.

2. Objectives

- **O1. Build a reproducible analysis cohort** from the pinned Open Play data (eligible players aged 18–40 with GDT at waves 1 and 6; 710 people before the telemetry-coverage filter).

- **O2. Engineer interpretable early-window features** (total time and frequency, daily irregularity, night-time share, long sessions, weekly pattern and trend, game variety) at a fixed prediction time of wave 1 + 14 days.
- **O3. Quantify added value (RQ1)** compare M0–M3 on the same people, folds, and tuning budget to test whether telemetry improves prediction of the wave 6 GDT, and therefore of GDT change, over the baseline GDT and total hours.
- **O4. Test the secondary target (RQ2)** predict next-survey interference with daily routines from the preceding 14 days of telemetry and the current interference score.
- **O5. Explain and communicate results** with SHAP and permutation importance, uncertainty intervals, and a small Streamlit research dashboard that abstains when data are insufficient.
- **O6. (Optional) Evaluate an NLP extension (RQ3)** coding baseline open-ended responses to test whether they add information (M4).

Together these objectives address the problem by replacing the assumption that more telemetry is better with an explicit, reproducible test of whether it helps.

3. Background

Problematic gaming affects a small but meaningful share of players, and the ICD-11 framework stresses that long playing hours alone do not constitute a disorder; marked functional impairment is required (World Health Organization, 2020). The GDT (Pontes et al., 2021) provides a short four-item measure of symptom burden on a 4–20 scale. Existing solutions are largely awareness tools, such as Android Digital Wellbeing and Apple Screen Time, which show usage and offer timers (Google, n.d.; Apple, n.d.). The documentation reviewed does not show that these tools are validated predictors of GDT change. Research examples include a two-wave machine-learning study using the player–avatar bond to predict later risk (Stavropoulos et al., 2023) and the Open Play infrastructure, which links multi-platform gameplay logs with psychological measures (Ballou et al., n.d.).

A machine learning approach is useful because the question involves many correlated, partly nonlinear behavioral signals (when, how regularly, and how long people play) whose joint contribution cannot be tested with a single correlation. It is also necessary for a fair test of added value: regularized and tree-based models can be compared against strong baselines under identical person-level cross-validation. We keep the scope modest. The GDT recall window of three months overlaps between waves, so the task is to predict symptom burden measured later, not disorder onset.

4. Methodology

Targets. The primary target is the wave 6 GDT total (and, by subtracting the baseline, the GDT change); the secondary target is interference with daily routines at the next survey, with the sign of gaming_value_routines reversed so that higher means more interference.

Models. Five models are compared on the same eligible people, folds, and tuning budget:

Model	Information used	Purpose
M0	Wave 6 GDT = baseline GDT	No-change control
M1	Baseline GDT only	Strong baseline learned from the score
M2	M1 + total weekly hours in first 14 days	Added value of playtime

M3	M2 + predefined behavioral features	Added value of timing and patterns
M4 (optional)	M3 + labels from baseline open-ended text	Added value of NLP

Algorithms. Ridge and Elastic Net are the primary interpretable baselines; Gradient Boosting is the main nonlinear candidate; Random Forest is a secondary comparison. Deep sequence models (LSTM, Transformer) are not used in the first version because only about 710 independent labels are available. All models are built as scikit-learn Pipelines so that imputation, scaling, and feature selection are learned inside each training fold.

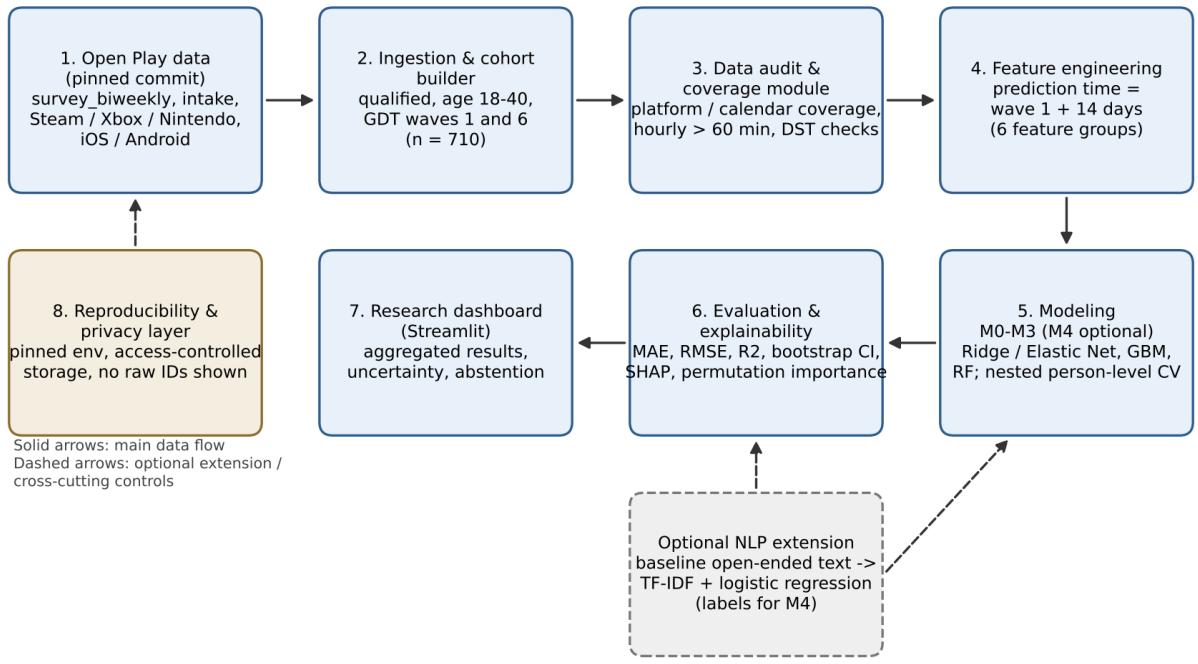
Validation and metrics. Person-level nested cross-validation (five outer, three inner folds) prevents different waves of the same person from appearing on both sides of a split. The primary metric is mean absolute error (MAE), with RMSE and R^2 as auxiliary metrics. The M3–M2 person-level error difference is reported with a bootstrap interval resampled over people. SHAP and permutation importance explain the final model on unseen outer-fold people. A small, uncertain gain will be reported as such; feature importance is not evidence of causality.

Optional NLP. Baseline texts can be labeled with multiple categories (for example, difficulty controlling, routine or sleep impact) using TF-IDF with regularized logistic regression; a language model would be added only as a candidate evaluated against human labels (scikit-learn, n.d.-d).

5. Architecture Design Diagram

Figure 1 shows the end-to-end research pipeline. Data flow from the pinned Open Play files through cohort construction, auditing, feature engineering, modeling, and evaluation to a small dashboard. A cross-cutting reproducibility and privacy layer applies to every stage.

Figure 1. High-level architecture of the Problematic Gaming Early Warning research pipeline



- **1. Open Play data** the pinned GitHub commit (eb0a68c1…) providing biweekly surveys, intake screening, and platform telemetry (Steam, Xbox, Nintendo Switch, iOS, Android). It is the only data source.
- **2. Ingestion and cohort builder** reads the compressed CSVs, joins survey to intake IDs, and applies eligibility rules (qualified = True, age 18–40, complete GDT at waves 1 and 6). Exclusions are logged in a flow table.
- **3. Data audit and coverage module** computes platform and calendar coverage per person, distinguishes missing records from non-play, and checks hourly totals above 60 minutes, day-crossing sessions, and daylight-saving transitions.
- **4. Feature engineering** builds six feature groups from the 14 days after wave 1 only, so nothing after the prediction time enters the inputs; coverage ratios are included as features.
- **5. Modeling** trains M0–M3 (and M4 if text coverage allows) inside person-level nested cross-validation.
- **6. Evaluation and explainability** reports MAE, RMSE, R^2, bootstrap intervals, sensitivity analyses (coverage thresholds, window length), SHAP, and permutation importance.
- **7. Research dashboard (Streamlit)** shows aggregated results, behavior changes, and model uncertainty, and abstains when data are insufficient. It is a research interface, not a real-time monitoring app.
- **8. Reproducibility and privacy layer** pinned dependencies and data version, access-controlled storage, minimal columns, and no person IDs or raw schedules in outputs.
- **Optional NLP extension** labels baseline open-ended text and feeds M4 and the evaluation stage.

6. Data Sources

The project uses Open Play (Ballou et al., n.d.), which combines behavioral logs with psychological measures from the same people. Calculations are pinned to GitHub commit eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840 (accessed 11 October 2026). The survey files provide the four GDT items (coded 1–5, summed to 4–20), the five effects-on-life items (recoded so that higher means more interference), and open-ended responses; the intake file provides eligibility, age, and time zone. Telemetry comes from Steam (hourly changes in total time), Xbox and Nintendo Switch (sessions), and iOS and Android (daily totals from screenshots), so platforms differ in resolution and not every feature exists everywhere. The preliminary cohort consists of 710 eligible people with both GDT measurements, a number that may shrink after the telemetry-coverage filter. Preprocessing includes cohort filtering, coverage computation, local-time conversion, audits of impossible durations, and per-fold imputation and scaling. Data are CC0 with a no-reidentification clause (Digital Wellbeing, n.d.-d), and only the required columns are kept.

7. Literature Review

The GDT was developed and validated as a short single-factor measure of WHO-framework gaming disorder symptoms (Pontes et al., 2021), giving a sound outcome variable. Self-reported and logged use agree only moderately (Parry et al., 2021), and in League of Legends players both time measures showed small associations with symptoms (Jin et al., 2022), which motivates logged time as a strong comparison model. Large-scale logged data from seven games found little evidence that average playtime affects general well-being (Vuorre et al., 2022), cautioning against equating time with harm,

and a two-wave machine-learning study predicted later gaming disorder risk mainly from self-reported avatar-bond variables (Stavropoulos et al., 2023). None of these studies tests whether detailed multi-platform telemetry from an early window adds predictive value beyond the baseline GDT; Ballou et al. (n.d.) name this as an open research opportunity. Our project builds on and extends this work by running that test openly, on identical people, splits, and tuning budgets, and without assuming the result in advance.

Implementation Plan

1. Technology Stack

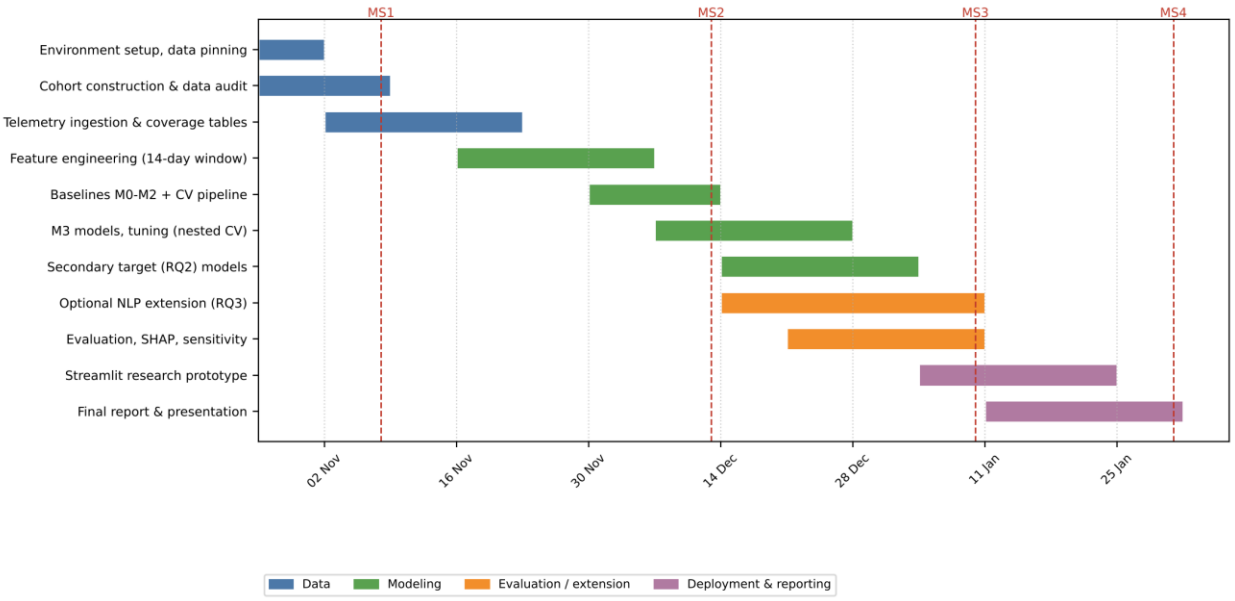
Layer	Choice	Use in the project
Language	Python 3.11	All data, modeling, and dashboard code
Data libraries	pandas, NumPy, pyarrow	Compressed CSV reading, date handling, feature generation
Modeling	scikit-learn (Pipeline, GroupKFold, Ridge, Elastic Net, Random Forest, GradientBoosting), optionally LightGBM	Person-level nested CV and model comparison
Explainability	SHAP, scikit-learn permutation importance	Explaining final-model output
NLP (optional)	scikit-learn TfidfVectorizer + LogisticRegression	Open-ended text labeling (M4)
Visualization	Matplotlib, Plotly	Descriptive plots and figures
Prototype / deployment	Streamlit	Research dashboard on aggregated results
Environment	Jupyter, Git/GitHub, pinned requirements.txt (or conda lock)	Reproducible scripts and notebooks, version control
Hardware	Standard laptop, CPU only	Tabular models on about 710 people need no GPU; no paid API required

Deployment is limited to a local Streamlit app. Because of the no-reidentification terms, the app uses aggregated results or synthetic examples only; any public hosting would use those only.

2. Timeline

The plan covers 26 October 2026 to 31 January 2027 (14 weeks). Dates should be adjusted to the official course deadlines if these differ.

Figure 2. Project timeline (26 Oct 2026 - 31 Jan 2027)



Phase	Task	Dates	Deliverable
Data collection & preprocessing	Environment setup, data pinning	26 Oct – 1 Nov	Repo, pinned environment
	Cohort construction & data audit	26 Oct – 8 Nov	Exclusion flow table, audit report
	Telemetry ingestion & coverage tables	2 – 22 Nov	Coverage per person/platform
Model development	Feature engineering (14-day window)	16 Nov – 6 Dec	Analysis table
	Baselines M0–M2 + CV pipeline	30 Nov – 13 Dec	Baseline results
Training & evaluation	M3 models, tuning (nested CV)	7 – 27 Dec	M3 vs M2 comparison
	Secondary target (RQ2) models	14 Dec – 3 Jan	RQ2 results
	Optional NLP extension (RQ3)	14 Dec – 10 Jan	Label guide, M4 results
	Evaluation, SHAP, sensitivity	21 Dec – 10 Jan	Final metrics, intervals
Deployment	Streamlit research prototype	4 – 24 Jan	Running dashboard
	Final report & presentation	11 – 31 Jan	Report, slides

Task distribution matrix (R = responsible, S = support):

Task	Yunus Berkay	Deniz	Fatih	İsa Adahan	Alperen
Environment, data pinning, repository		R		S	
Cohort construction & data audit		S	R		
Telemetry ingestion & coverage tables	S				R
Feature engineering		S		R	
Baselines M0–M2 + CV pipeline		R			S
M3 models and tuning	R	S			
Secondary target (RQ2) models		S			R
Optional NLP extension (labeling, TF-IDF)		S		R	

Evaluation, SHAP, sensitivity analysis	R		S		
Streamlit prototype	S		R		
Final report & presentation	R	R	R	R	R

Assignments were drawn at random; each task has one responsible member (R) and one supporting member (S), and everyone shares the final report.

3. Milestones

#	Milestone	Date	Completion criterion
MS1	Cohort and data audit complete	8 Nov 2026	Exclusion flow and 710-person cohort reproduced from the pinned commit
MS2	Analysis table and baselines ready	13 Dec 2026	Feature table built with the 14-day boundary; M0–M2 evaluated; coverage threshold fixed before looking at results
MS3	Models evaluated and explained	10 Jan 2027	M3 vs M2 error difference with bootstrap interval; SHAP outputs; RQ2 results
MS4	Final delivery	31 Jan 2027	Working Streamlit prototype, final report, and presentation

4. Challenges and Mitigations

Area	Challenge	Mitigation
Data quality	Telemetry is missing for many people; platforms differ in resolution; missing records can be confused with non-play; hourly totals above 60 minutes and DST errors	Record coverage ratios; never code missing as zero; fix the minimum coverage rule (for example 80%, with 60% and 90% as sensitivity checks) before seeing results; audit durations and time conversion
Data quality	Attrition: only 710 of 1,598 baseline participants have both GDT measurements, so those lost may differ	Report the exclusion flow and compare completers with non-completers; no claims beyond the matched cohort
Model performance	Small sample (at most 710 people) and noisy target; telemetry may add little	Few interpretable features, regularization, person-level nested CV; report null or small gains honestly, as an informative result
Model performance	Overlapping GDT recall windows make 'future' symptoms hard to separate	State the limitation; frame the task as predicting later symptom burden; use RQ2 as the more forward-looking test
Technical constraints	Leakage risk across waves or from post-prediction data	GroupKFold by person; fit all preprocessing inside folds; fix inputs to the prediction time
Technical constraints	Limited compute, data size, and time	CPU-friendly models; precompute daily per-person features; keep the NLP part optional

5. Ethical Considerations

Data privacy. Open Play is shared under CC0 with a no-reidentification clause; open access does not mean the data are fully anonymous. We will not try to identify anyone or link identities to gaming accounts or external profiles (Digital Wellbeing, n.d.-d). Only necessary columns are kept (political views, detailed geography, and unneeded health fields are excluded), raw data access is limited to the team and stored in an access-controlled place, and no person IDs, raw daily schedules, or recognizable open-

ended texts appear in any output. This follows the data minimization principle (Personal Data Protection Authority, n.d.). No new participant data are collected; a future user study in Türkiye would need separate ethical review, a legal basis, and explicit consent.

Bias and fairness. The sample is voluntary, drawn from US and UK players aged 18–40, with unequal platform coverage and non-random attrition, so results cannot be generalized to students in Türkiye, children, or people seeking clinical help. We will report performance by platform-coverage group where sample size allows and avoid using sensitive attributes as predictors.

Impact on the target community. Predictions are scale-based research outputs and not diagnoses. Because feature importance does not prove causality, the interface will not recommend treatment, will show uncertainty, and will abstain when data are insufficient. It will avoid streaks, rewards, or constant alerts and allow users to turn off notifications and delete data. A risk of misuse (for example, by employers or schools) is reduced by not building individual-level monitoring in this project. For support, users in Türkiye can be referred to YEDAM and the 115 line (Yeşilay Counseling Center, n.d.).

6. References

Apple. (n.d.). Use Screen Time to manage your child's iPhone or iPad. Apple Support. https://support.apple.com/en-us/108806

Ballou, N., Földes, T. A., Vuorre, M., Hakman, T., Magnusson, K., & Przybylski, A. K. (n.d.). Open Play: A longitudinal dataset of multi-platform video game digital trace data and psychological measures [Preprint and online data documentation]. https://digital-wellbeing.github.io/open-play/

Digital Wellbeing. (n.d.-a). Open Play [Data and code repository; version eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840]. GitHub. https://github.com/digital-wellbeing/open-play/tree/eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840

Digital Wellbeing. (n.d.-d). CC0 1.0 Universal with No-Reidentification Clause [Data terms of use]. GitHub. https://github.com/digital-wellbeing/open-play/blob/eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840/LICENSE

Google. (n.d.). Manage how you spend time on your Android phone with Digital Wellbeing. Android Help. https://support.google.com/android/answer/9346420

Jin, J. T., Kittaneh, A. A., Sidhu, N. K., & Lechner, W. V. (2022). Incorporating objective behavioral data in gaming disorder research: Associations between time spent gaming and gaming disorder symptoms. Computers in Human Behavior, 136, 107378. https://doi.org/10.1016/j.chb.2022.107378

Parry, D. A., Davidson, B. I., Sewall, C. J. R., Fisher, J. T., Mieczkowski, H., & Quintana, D. S. (2021). A systematic review and meta-analysis of discrepancies between logged and self-reported digital media use. Nature Human Behaviour, 5, 1535–1547. https://doi.org/10.1038/s41562-021-01117-5

Personal Data Protection Authority (Kişisel Verileri Koruma Kurumu). (n.d.). Basic principles for the processing of personal data [Kişisel verilerin işlenmesine ilişkin temel ilkeler]. https://www.kvkk.gov.tr/Icerik/4189/Kisisel-Verilerin-Islenmesine-Iliskin-Temel-Ilkeler

Pontes, H. M., Schivinski, B., Sindermann, C., Li, M., Becker, B., Zhou, M., & Montag, C. (2021). Measurement and conceptualization of gaming disorder according to the World Health Organization framework: The development of the Gaming Disorder Test. International Journal of Mental Health and Addiction, 19, 508–528. https://doi.org/10.1007/s11469-019-00088-z

scikit-learn. (n.d.-d). Feature extraction [Software documentation]. https://scikit-learn.org/stable/modules/feature_extraction.html

Stavropoulos, V., Zarate, D., Prokofieva, M., Van de Berg, N., Karimi, L., Gorman Alesi, A., Richards, M., Bennet, S., & Griffiths, M. D. (2023). Deep learning(s) in gaming disorder through the user-avatar bond: A longitudinal study using machine learning. Journal of Behavioral Addictions, 12(4), 878–894. https://doi.org/10.1556/2006.2023.00062

Vuorre, M., Johannes, N., Magnusson, K., & Przybylski, A. K. (2022). Time spent playing video games is unlikely to impact well-being. Royal Society Open Science, 9, 220411. https://doi.org/10.1098/rsos.220411

World Health Organization. (2020, October 22). Addictive behaviours: Gaming disorder. https://www.who.int/news-room/questions-and-answers/item/addictive-behaviours-gaming-disorder

Yeşilay Counseling Center (Yeşilay Danışmanlık Merkezi). (n.d.). YEDAM. https://www.yedam.org.tr/